



# Image Denoising by Wavelet Transform Based on New Threshold

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**Abstract.** On the basis of two classical hard and soft threshold processing methods, combined with the improved methods mentioned in the literature, a comprehensive threshold processing method is proposed. The threshold function can not only overcome the defects of discontinuity and constant deviation of the traditional threshold, but also adjust the error between the original wavelet coefficient and the thresholding coefficient by adjusting the parameters. Through the comparison of simulation experiments, it is found that the denoising effect of the new threshold function is significantly improved over the traditional threshold denoising effect in terms of visual effect, mean square error, peak signal-to-noise ratio, etc.

**Keywords:** Wavelet analysis · Threshold · Image denoising

## 1 Introduction

With the Great advances of modern science and strength, the network is everywhere. People get a lot of information through cell phones, computers and so on. Pictures are a major tool for information dissemination. In the process of image transmission, conversion and reception, it will inevitably be polluted by various kinds of noise. However, for example, edge detection, feature extraction, pattern recognition and so on, usually need effective denoising algorithm to preprocess, so as to get a more real image. Therefore, how to remove the image noise is a topic worthy of study. It is the key to remove the noise without destroying the contrast, definition and texture information of the image while suppressing the noise. Using wavelet transform to remove the noise of the image is a common method.

Wavelet analysis is a very popular frontier in the world in recent years. It is a breakthrough after Fourier analysis. It has brought brand-new ideas to many related fields and provided powerful tools. From an engineering point of view, Shortwave analysis is a Signal analysis and processing method and an helpful method of analyzing time frequency band after Fourier's Transformation. As a new multi-resolution analysis method, wavelet transform can analyze both time domain and frequency domain. It has a characteristics of Time frequency detail, localization and MultiRes, Therefore, it is especially suitable for processing unstable signals. Different from single wavelet, multi-

wavelet bases are generated by multi-wavelet generating functions, corresponding to multi-scale functions. Single wavelets can directly decompose and reconstruct the sampled data when processing the signal and signal, but multi wavelets can not. They have to do Data pre-processing before decomposition, then decompose the processed data and reconstruct the reconstructed data, the reconstructed data can be obtained only by line post-processing. Multiwavelets have more degrees of freedom when constructed, so they have shorter branches, shorter sets, and more vanishing moments than single wavelets, and they can satisfy both orthogonality and symmetry. Multi-wavelet not only keeps many advantages of single wavelet, and overcame the defects of a single small wave. In practice, the important smoothing, compactness and symmetry can be combined perfectly.

Wavelet transform threshold hold-up method was first put forward by D.L.donoho in 1995. The method has been widely used in hard and soft threshold functions [9, 10], but it also has some shortcomings. Although the two threshold functions proposed in [1, 2] have improved the solid threshold and mild threshold functions to a certain extent, this effect is non particularly obvious, and reconstructed image has poor error control in details.

This article offers a new enhanced threshold feature that combines the pros and cons of flexible and hard threshold functions [3, 4]. The expression of the threshold function is simple, and it is the extension of the existing threshold function. The system error between the threshold coefficient and the original coefficient is adjusted by adjusting the parameters. At the same time, it is also differentiable and easy to calculate.

## 2 Wavelet Threshold Denoising Principle

The first method of reducing threshold noise is the visual reduction method [8] proposed by Donoghue. Gasoline is to convert the noise signal into a corresponding small wave factor by small wave transformation. Set an appropriate threshold value for the signal, So when the threshold value exceeds the wavelet transform coefficient, we think that the signal is caused by noise and contains no information component, so we discard the signal; When the threshold value is less than the wavelet transform coefficient, we think that the signal is the result of the signal itself and noise, and keep the signal (solid threshold means) or Zoom to zero according to a certain Fixed value(mild threshold means).Finally, use small wave transformation to rebuild the new factor and get the signal after the noise.

The main steps are as below: ① select the appropriate level  $N$  of wavelet decomposition , and make discrete wavelet transform for function  $f(j, l)$  to obtain the wavelet coefficients  $w(j, l)$  ; ② set the threshold value of each decomposition layer, process the threshold value of each layer of wavelet coefficient, and calculate the wavelet estimation coefficient  $\hat{w}(j, l)$ ; ③ reconstruct the wavelet coefficient, calculate and reconstruct the image through the  $n$ th layer of low-frequency band coefficient  $\hat{w}(j, l)$  of wavelet decomposition and the high-frequency coefficient of each layer after quantization, and obtain the denoised image  $\hat{f}(j, l)$

### 3 Wavelet Threshold Function

#### 3.1 Commonly Used Wavelet Threshold Function

##### (1) solid threshold

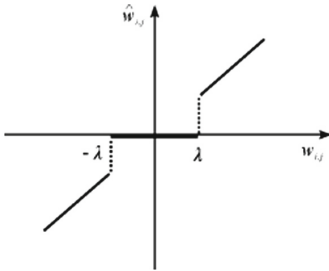
If the modulus of the small wavelet factor  $w$  is smaller than a designated threshold  $\lambda$ , set up to 0, and if it is greater than the threshold  $\lambda$ , it remains invariant, that is:

$$w_{\lambda} = \begin{cases} w & |w| \geq \lambda \\ 0 & |w| < \lambda \end{cases} \quad (1)$$

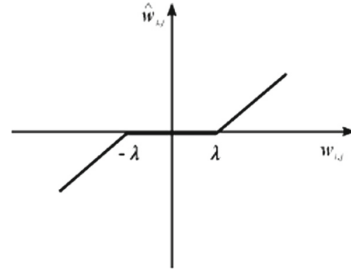
##### mild threshold

If the modulus of the wavelet factor  $w$  is smaller than the designated threshold  $\lambda$ , it is set to 0, and if it is greater than the threshold  $\lambda$ , it is set to subtract the threshold, that is (Figs. 1 and 2):

$$w_{\lambda} = \begin{cases} [sign(w)](|w| - \lambda) & |w| \geq \lambda \\ 0 & |w| < \lambda \end{cases} \quad (2)$$



**Fig. 1.** The hard thresholding



**Fig. 2.** Thesoft thresholding

Although solid thresholding function and mild thresholding function are widely used in practice, they also have some defects: for example, when the hard thresholding function is used to process the image, the continuity of the estimated wavelet coefficients is poor, which may cause the reconstructed image to produce visual distortion such as oscillation and Gibbs effect; while the wavelet factor evaluated by the mild thresholding function are generally continuous. However, when the wavelet coefficients are large, The constant discrepancy between the estimated small wave factor and the original small wave factor affects the approximation between the reconstructed image and the actual image, resulting in an inevitable error in image reconstruction. In addition, the derivative of traditional soft threshold function is discontinuous, but in practice, the higher derivative is often treated, so it has certain limitations [8–10].

### 3.2 Streamlined Wavelet Shrinkage Function

According to the advantages of the above threshold function and the methods mentioned in [1, 2], A new wavelet shrinkage has been advised to estimate small wave factor, whose functions are as follows:

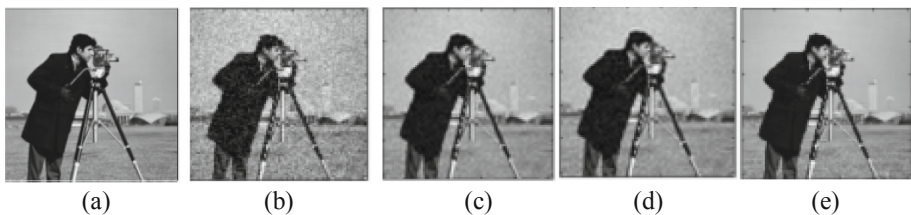
$$\hat{w}_\lambda = \begin{cases} w & |w| \geq \lambda_1 \\ w(1 - \frac{|\lambda_2|}{|w|})^n & \lambda_2 \leq |w| < \lambda_1 \\ 0 & |w| < \lambda_2 \end{cases} \quad (3)$$

Where  $w$  is the wavelet coefficient,  $\hat{w}_\lambda$  is the estimated wavelet coefficient,  $\lambda_1 = \sigma\sqrt{2 \ln N + 2 \ln \ln N}$ ,

$\lambda_2 = \alpha\sigma\sqrt{2 \ln N + 2 \ln \ln N}$  [5-7].  $\alpha$  and  $n$  is an adjustable parameter,  $0 \leq \alpha \leq 1$ ,  $n$  is a positive integer. When the wavelet coefficient  $\lambda_1$  is larger than  $m$ , the coefficient is invariable, When the wavelet coefficient is less than  $\lambda_2$ , let it be 0, When the wavelet coefficients are between  $\lambda_1$  and  $\lambda_2$ , a new threshold function is used.

## 4 Simulation Results and Analysis

For the sake of checking the dehumidification effect of new shrinkage function, article selects the solid thresholding function and the mild thresholding function to compare them, In the experiment, the noise of  $\sigma = 10$  is added to the original image first, Three-layer decomposition of noisy image by selecting wavelet basis sym4, In the simulation test, two thresholds  $\lambda_1 = \sigma\sqrt{2 \ln N + 2 \ln \ln N}$ ,  $\lambda_2 = \alpha\sigma\sqrt{2 \ln N + 2 \ln \ln N}$ , Wherein  $\sigma$  is the standard deviation and the total number of pixels in the image are expressed by  $n$ ,  $\alpha = 0.5$ , and the parameter can be adjusted by  $n = 2$ , Fig. 3 is the experimental result of the cameraman image with the size of  $256 * 256$  pixels. It can be seen from figure (e) that the noise content of the image is significantly reduced, and the texture features of the image can be effectively preserved.



**Fig. 3.** Comparison of denoising effect of each threshold function

(a)-The original image (b)-The Noisy image (c)-The Hard threshold denoising (d)-The Soft threshold denoising (e)-The Improved threshold denoising

For the sake of better proving the noise-reduction effect of the new shrinkage function proposed in the article, we use the Uniform Error and Maximum Noise Report

(PSNR) to analyze the image noise reduction effect. The lower the average error value, the better the picture quality. The taller the PSNR value, the greater the picture quality.

$$PSNR = 10 \log_{10} \left( \frac{255^2}{MSE} \right) \quad (4)$$

$$MSE = \frac{1}{M} \times \frac{1}{N} \times \sum_{i=1}^M \sum_{j=1}^N [f'(i,j) - f(i,j)]^2 \quad (5)$$

Analysis Table 1 shows that the new shrinkage function denoising method put forward in the article has greatly improved the signal-to-noise ratio SNR, while the mean square error MSE has also significantly decreased, It helps to better preserve the contours and details of the image. and the denoising effect is very good.

**Table 1.** Comparison of denoising results of cameraman image

|     | Hard threshold denoisin | Soft threshold denoising | Improved threshold denoising |
|-----|-------------------------|--------------------------|------------------------------|
| SNR | 28.3545                 | 28.5821                  | 29.6684                      |
| MSE | 0.1214                  | 0.1121                   | 0.0972                       |

## 5 Conclusion

The traditional solid and the mild thresholding function and the two shrinkage functions proposed in [1, 2] can not effectively express the energy distribution, so it is necessary to find a balance between denoising and retaining edge information. The revised thresholding function proposed in this article overcomes the shortcomings of the existing thresholding function, such as the deviation of image denoising, the blur of image or the unclear texture or the blur of edge after denoising, and energy adaptability. The results of the simulation indicate that the method can, to some extent, obtain lower average errors and better image details. Compared with the conventional shrinkag function and the improved shrinkag function, it improves the visual effect and signal-to-noise ratio of the noise suppression image.

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